ANDERSON SERANGOON JUNIOR COLLEGE

JC2 H1 Geography Preliminary Exam (2025)

H1 GEOGRAPHY **8834/ 01**

25 August 2025

3 hours

Additional Materials: Answer Booklet
1 Insert

READ THESE INSTRUCTIONS FIRST
If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your name and class on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer Question 1 in **Section A**.
Answer both questions in **Section B**.
Answer **two** questions in **Section C**.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Cluster 2 Fieldwork

1 Eight students conducted a study on a Saturday morning in June (9:00–11:00 AM) to investigate whether frequent visits to Kim Pong Park improve the mental well-being of elderly residents. They located their fieldwork investigation in a cluster of eight low-rise apartment blocks in Tiong Bahru. Light rain had fallen earlier in the morning before the study.

They agreed on the following hypothesis:

"Elderly residents who often visit neighbourhood parks tend to have better mental well-being."

The students split themselves into two groups and surveyed 20 elderly residents:

- Group 1 (4 students): Conducted on-the-spot interviews at the park exit with 10 elderly visitors.
- Group 2 (4 students): Conducted in-home interviews with 10 residents from Blocks 49 and 50.

Resource 1 shows Kim Pong Park with an exit point and nearby low-rise apartment blocks (43 to 50). Resource 2 shows the survey questionnaire used by the students. Resource 3 shows data collected on the park visit frequency and self-reported mood among the elderly participants.

(a) Explain whether the students' hypothesis is suitable. [3]

- The study site is small (a cluster of apartment blocks, one park) which is manageable for a small group of 8 students to cover in a limited time (2 hours).
- The hypothesis is clear and testable and could be investigated by collecting primary data through questionnaire surveys of 20 respondents.
- The hypothesis is relevant to geography/social studies and directly addresses the relationship between frequency of park visits (independent variable) and mental well-being (dependent variable) in elderly respondents

(b) With reference to Resource 1, explain how each group of students could have best selected the respondents for their interviews. [4]

Before setting off, define the target population — for example, specify eligibility as individuals aged 65 years and above.

Group 1 (Park Exit Interviews):

• Systematic sampling will be employed by selecting every nth elderly person exiting the park. The value of n will be determined from on-site observations of visitor flow to ensure a total of 10 interviews is achieved within the 2-hour period. For example, every 2nd or 3rd elderly visitor who leaves the park). They could monitor the flow rate and adjust the interval in real-time to ensure the quota is met by the end of the study period.

Group 2 (In-home Interviews):

Ensure both blocks are represented, aim to select 5 respondents from each block. If one block has more elderly than the other, allocate proportionally (e.g., Block A 60% of elderly → 6 interviews; Block B 40% → 4 interviews).

In each block (4 stories): Systematically knock on doors that meet the age criterion (e.g., every 3rd or 4th apartment door on each floor)

If difficulties arise in finding the last few respondents, snowball sampling will be employed as a supplementary strategy. After each interview, participants will be asked whether they know neighbours in the block who meet the age criterion and may be willing to participate.

(c) With reference to Resource 1, suggest how the students could minimise potential risks in conducting the survey. [3]

Potential risks	How to minimize risks
Personal safety risks: Students may face harassment, theft, or physical harm, especially if interviewing strangers in secluded or poorly lit locations. Students may get lost or separated from groups in the estate or park.	Conduct a site-specific risk assessment before fieldwork, considering factors such as hidden corners and lighting conditions. Always operate in pairs or small teams to enhance personal safety and provide witnesses in case of incidents. Hold interviews in well-lit, busy areas such as parks or communal spaces in residential blocks, avoiding secluded spots, stairwells, or other isolated locations. Share fieldwork plans and expected timelines with teachers and check in at set intervals to ensure safety.
Emotional and psychological risks: Sensitive questions may distress the elderly participants, potentially leading to emotional upset or conflict.	Avoid sensitive or intrusive questioning. Be ready to pause or terminate interviews if participants feel distressed. At the conclusion of the interview, participants will be thanked for their time and reassured that all information they have shared will remain strictly confidential.
Health and environmental risks: Exposure to rain, heat. Insect bites or slippery/uneven surfaces. Fatigue or dehydration.	Check the weather forecast and schedule interviews under safe conditions. Wear suitable clothing and footwear, and bring insect repellent, hats, sunscreen, water, and umbrellas as needed. Take regular breaks to prevent fatigue and ensure all students stay hydrated. Pay attention to uneven paths or slippery surfaces, especially if it has recently rained.

(d) Explain the strengths and limitations of the survey questionnaire shown in Resource 2. [5]

<u>Strengths</u>

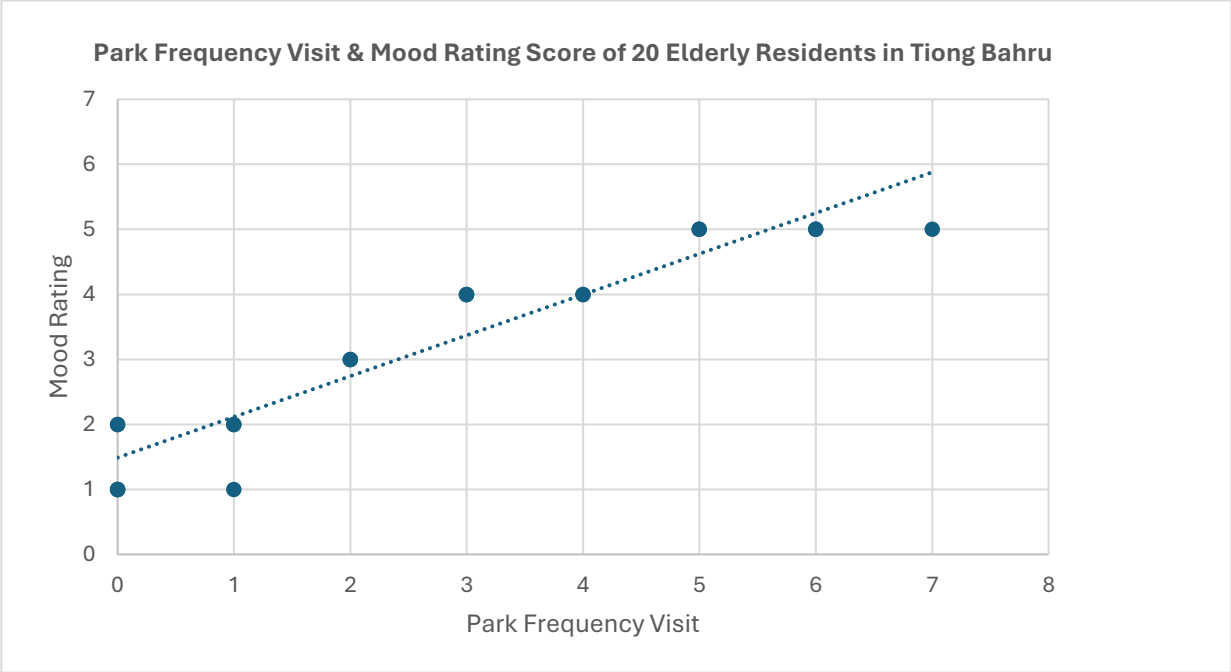
Using short, mostly close-ended questions and limiting the total number of questions helps elderly participants respond more easily and accurately, reducing fatigue and improving the reliability of the data.

Using a colour gradient chart to reflect mood levels can be particularly helpful for elderly participants as it provides immediate visual clarity. Colours can convey information quickly and intuitively. For example, a gradient from red (low mood) to green (high mood) makes it easy to see differences at a glance. Elderly participants may find visual cues easier to interpret than numerical scales or text alone.

<u>Limitations: Problematic questions in the questionnaire</u>

Q.5 is a leading question: By asking “Don’t you agree…?” the question pushes respondents toward answering “yes,” even if they might not fully agree. It contains an assumption – The question assumes that parks are indeed the best way for seniors to feel less depressed, which may not be true for all participants.

Q.6 is a judgmental and potentially embarrassing question that uses highly negative language ("lonely," "nothing else to do"). Respondents may not tell the truth to avoid admitting to these feelings.

		Q.7 The term "old people" is often considered outdated and potentially offensive. It's best to use "seniors," "older adults," "the elderly," or "people aged X and over."	
	(e)	Sketch a graph to present the information on “Visits per week” and “Mood Rating” in Resource 3 and explain its suitability.	[5]
		For full marks, the scatter plot must include a descriptive title, clearly labeled X and Y axes with appropriate units, accurately plotted data points, and a trend line illustrating the data's relationship.  	
	(f)	Evaluate the validity of the students’ investigation. Several aspects of the investigation enhance the validity of its conclusions. First, the study employed quantifiable measures of park visitation (visits per week), paired with mood rating scale (1-5), that were directly recorded, giving comparable data across respondents. Consistent questioning, where all respondents were asked the same questions, further reduced potential interviewer bias and improved reliability. Importantly, the link between park visits and well-being is a meaningful one, supported by existing literature on nature and health. Conducting face-to-face interviews in the elderly’s homes allowed participants to respond in a familiar and comfortable setting, while conducting the study in a real park setting also enhances ecological validity. Collecting qualitative data on reasons for visiting (exercising, relaxation etc.) added depth and insight beyond just numbers to the findings. Finally, the emergence of consistent trends as shown in the scatter plot (e.g., higher visit frequency aligning with higher well-being scores) provide further evidence supporting the conclusions. However, several limitations reduce the strength of the conclusions. The small sample size (20), collected over a single Saturday morning for two hours, does not represent the wider elderly population of Tiong Bahru. Only two blocks nearest the park were selected, introducing sampling bias and likely over-representing frequent park visitors, as residents living farther away may visit less often. Treating all elderly as a single category overlooks differences between Young Seniors (65–74) and Old Seniors (75+), potentially masking real variations. Young Seniors (the “Third Age”) are generally healthier, more active, and often report higher well-being after retirement. Combining their results with those of Old Seniors (the “Fourth Age”) inflates the overall average, concealing the challenges faced by the older group. Consequently, the study risks drawing	[10]

		misleading conclusions, such as suggesting that “elderly mental well-being is high” or that “age has no significant impact,” neither of which reflects reality. Furthermore, insensitive or poorly worded questions may trigger emotional responses unrelated to the concepts being measured or bias answers, reducing the survey’s ability to accurately assess mental well-being. Although the hypothesis assumed, and the survey suggested, that park visits improve mental well-being, this relationship may reflect correlation rather than causation. Other factors, such as strong social networks or physical health, could influence both park visits and well-being. For example, an elderly person who is socially active may have higher well-being regardless of park visits, while someone with limited mobility may visit less but experience lower well-being due to health issues rather than lack of park access. Overall, while the study gives some useful indications of the relationship between park visits and well-being, the findings should be interpreted with caution.	

Section B

Cluster 1 Sustainable Future and Climate Change

Answer **both** questions

2	Resource 4 shows the percentage share of urban population living in slums for select regions from 1990 to 2014. Resource 5 shows the percentage share of urban population living in slums in Africa in 2014. Resource 6A shows a photograph of the Makoko informal settlement in Lagos, Nigeria. Resource 6B shows a photograph of the Kibera informal settlement in Nairobi, Kenya.		
	(a)	With reference to Resource 4, compare the percentage of urban slum population between Sub-Saharan Africa and South Asia from 1990 to 2014.	[5]
		• Both experienced overall decrease → SSA decreased from 70% percentage share of urban population to 55% while SA decreased from 57% to 31% • Both consistently had higher % share of urban slum population than the global average → For all years, SSA was at least 25% higher than the global average, while SA was at least 2% higher • SA had larger overall decrease in percentage share of urban population compared to SSA → 26% for SA compared to 15% for SSA • SA had a constant rate of decrease (consistently about 5% decrease between periods), but SSA had a significant decrease in 2014 (decrease of about 6% in 2014, which is not consistent with the usual decrease of about 2% between periods) • Difference in % share of urban slum populations between SSA and SA was more drastic in later years → 13% difference in 1990 compared to 25% difference in 2014.	
	(b)	Describe the variations in the percentage share of urban population living in slums across Africa as seen in Resource 5.	[4]
		• Generally, countries in the Central and Eastern part of Africa had the highest percentage share of urban population living in slums, of more than 70%. • Countries in the South and North of Africa also largely had the lowest percentage share of urban population living in slums, of below 40%. • [Alternative] Generally, the countries near the equator had the highest percentage share of urban population living in slums, of more than 70%. • [Alternative] The percentage share of urban population living in slums decreased further North and South of the Equator, decreasing to less than 40% at the Northern and Southern parts of Africa. • However, Madagascar and Mozambique had high percentage share of urban population living in slums despite being located in the Southern part of Africa, of more than 70%. • Western part of Africa had moderate percentage share of urban population living in slums, of about 40-70%.	
	(c)	With reference to Resources 6A and 6B, suggest how the multiple deprivations experienced by slum dwellers may impact sustainable urban development.	[6]
		• [Environment] Presence of waste and litter in the waterbody as seen in Resource 6A suggests that the slum dwellers have a lack of access to proper	

		waste disposal and management systems. → This leads to water pollution and ecosystem degradation, reducing water quality and biodiversity. In the long run, this reduces the city's ability to maintain a healthy, liveable environment for current and future generations. • [Social] As seen in both Resources 6A and 6B, the dwellings are all located very close to each other, suggesting that the slum dwellers lack sufficient personal space and privacy. → High density increases the risk of disease outbreaks (e.g. respiratory illnesses, COVID-19) and creates unsafe, stressful living conditions, leading to lower quality of life and deepening inequality. This makes it harder to achieve the social inclusiveness and improved standards of living required for SUD. • [Social] Both Resources 6A and 6B shows the slum dwellings being constructed from make-shift and scrap materials which are not strong and easily flammable, suggesting that the slum dwellers lack access to safe and durable shelter. → Increases risks of injury, displacement, and loss of property. This weakens community resilience and strains public resources during disasters. • [Economic] Resource 6B shows some slum dwellers selling products at the entrance to their dwellings, suggesting that some slum dwellers lack access to formal jobs. → Informal work generates unstable incomes and does not contribute significantly to municipal tax revenues. This limits the city's ability to invest in urban infrastructure, education, and green initiatives, thereby restricting inclusive economic growth.	
	(d)	Explain why slums may develop in urban areas of less developed countries.	[4]
		• High rates of urban population growth & role of migration ○ Demand for housing often far outweighs the available supply of formal housing units as high rates of R-U migration persist due to significant push-pull factors that exist between rural and urban areas → e.g. job prospects; provision of social amenities, etc. • Urban poverty and lack of affordability of housing ○ Decrease in formal employment opportunities ○ Higher cost of living in urban centres ○ Commercialisation/privatisation of goods and services • Failure of governance to provide affordable housing ○ Inability to respond to rapid urbanisation which occurs at a pace that far exceeds the planning process to incorporate new migrants into the city. ○ General inertia on the part of urban authorities in taking steps to tackle the illegal occupation of urban lands, often due to the scale, sensitivity, and/or complexity of the issue → their sheer magnitude overwhelms the capacity/capability of existing programmes or policies to effectively address the issue ○ Lack of acknowledgement of rights of urban poor → Governments often fail to incorporate them into urban planning, thereby contributing to the growth of slums	
	(e)	Explain how the lives of slum dwellers in cities can be improved.	[3]
		• Explain strategy(s) to improve the lives of slum dwellers ○ Improvements to existing standards of living (Physical and/or non-physical)	

		○ Resettlement ○ Large-scale public housing ○ Sites-and-services scheme • Provide case-study(s) or example(s)	

3	Resource 7 shows the annual mean sea level trend (mm/year) in the Western Pacific region from 1993 to 2024. Resource 8 shows the relationship between adaptative capacity index and exposure index of countries at different levels of development. Resource 9A and 9B shows two different coastal adaptation strategies in Fiji.		
	(a)	With reference to Resource 7, describe the annual sea level trend in the western Pacific Ocean from 1993 to 2024.	[4]
		• General rise in mean sea level (MSL) of about 3-4mm/yr • Highest increase can be found in the Southeast sector with values 9-10mm/yr and SE coast of Australia. • Lowest values are recorded at latitude 35° and longitude 120°	
	(b)	Explain how changes in the extent of ice sheets can influence contemporary climate change.	[4]
		• Ice sheets have a high albedo, meaning they reflect a large portion of incoming solar radiation back into space. • When ice sheets shrink, more dark ocean or land surfaces are exposed, which absorb more heat, amplifying regional and global warming (a positive feedback).	
	(c)	With reference to Resource 8, describe the relationship between the adaptive capacity index and the exposure index of countries in relation to the different levels of development.	[4]
		The relationship between development level and adaptive capacity (AC) and exposure index (EI) is generally positive. As development level increases, AC increase but EI decreases while at low level of development, there is low AC but higher EI. • Advanced economies: Generally high AC (0.7-0.8) and often lower EI (0.17-0.35). • Emerging markets: AC is moderate (0.4-0.6) but EI varies widely (o.22-0.45). • Low-income countries: have lowest AC (0.15-0.5) and highest EI (0.25-0.55). This combination leads to their highest vulnerability. • Anomaly: The higher the AC, the higher the EI (countries such as Australia, Japan, Singapore)	
	(d)	Cite examples to explain why different groups of people have differences in climate change vulnerability. • Low-income communities who live in high-risk areas (landless farmers on floodplains, deltas, slum dwellers along riverbanks). Live in substandard housing, the poor cannot afford adaptive measures such as insulation, air conditioning, home repairs after disasters, no insurance payouts). Lack access to services (healthcare, emergency, evacuation and early warning during a climate hazard event). Landless residents who occupy marginal lands slum dwellers along river banks which are prone to river flood, landless farmers in coastal deltas prone to tropical cyclones (storm surge, powerful winds) • Agricultural and outdoor workers: Those who rely on the land and sea for their livelihood such as fishermen and farmers who have high exposure to extreme temperatures. May also suffer from loss of income and livelihood from climate-damage crops due to saltwater intrusion, drought and for fishermen from ocean acidification. In Bangladesh, rising seas are displacing rural communities , pushing them into already crowded urban slums. The Horn of Africa drought (2020-2023) led to severe food shortages affecting over 36million people,	[4]

especially pastoralists and subsistence farmers. Construction workers and miners also highly exposed to extreme heat and intense rainfall.

- Elderly populations: may have limited mobility, chronic health conditions, and reduced access to information or support during extreme weather events. The 2022 European hetwave caused over 60,000 deaths disproportionately affecting the elderly and those in poorly ventilated housing.

- Ethnic minority/indigenous groups. Black Americans during Hurricane Katrina. The 2019-2020 Australian bushfires disproportionately affected rural and indigenous communities (Aboriginals) , with long term health and economic impacts.

(e) Using Resources 9A and 9B, compare the impacts of the two coastal adaptation strategies on the environment and local communities. [6]

Mangrove restoration (Soft engineering, nature-based solution)		**Seawall** (Hard engineering solution)
▪ Both help dissipate the energy of sea waves and storm surges during a tropical cyclone, potentially protecting coastal communities from flooding and erosion. ▪ Both restrict residents' access to the sea.		
• Slows waves and reduces storm surge energy naturally, but takes years to mature for full protection. • Effectiveness may be reduced by extreme events before maturity.	Coastal protection effectiveness	• Provides immediate, high-strength protection against wave action and storm surges. • Performance may decline over time e.g. saltwater corrosion without costly maintenance.
• Restores natural habitats for fish, crustaceans, and birds. • Improves water quality by trapping sediments and pollutants. • Carbon sequestration.	Environmental Impact	• Alters natural coastal processes and sediment transport. No habitat benefits; may harm marine ecosystems due to wave reflection and scouring.
• Supports fisheries and ecotourism, providing long-term income opportunities. • Maintains natural coastal scenery and recreational value. • Requires community involvement for planting and monitoring.	**Impact on local communities**	• Protects property and infrastructure in high-value coastal zones. • Can reduce beach access and aesthetic value. • No direct livelihood benefits.

Section C

Answer **two** questions

4	Evaluate the issues faced by women living in the city.	[13]
	Possible Approaches: Candidates could approach the question by making a judgement on whether some issues are more significant than others through a consideration of their relative severity, prevalence, and impacts on women. The issues could be compared in relation to criteria such as their effects on women's economic well-being (e.g. wage gaps, limited employment opportunities, job insecurity), social well-being (e.g. unequal access to education, health services, or housing), and psychological well-being (e.g. safety concerns, discrimination, or mental health challenges). Candidates could also approach the question by making a judgement on whether women face similar or different issues in different urban contexts through a consideration of two or more case studies. Candidates could evaluate whether the issues are more acute in some cities than in others due to factors such as levels of development, cultural norms, political priorities, or economic structures. Candidates could also analyse whether some issues are universal to women living in cities, such as personal safety, while others may be context-specific, such as barriers to employment in certain industries. Candidates could further explore the scale of the issues (individual, community, societal) and the stakeholders involved (government, NGOs, community groups, private sector) that shape women's lived experiences in the city. *Levels marked using Generic Level Descriptors for H1 essays*	
5	'Less developed countries experience more severe impacts from contemporary climate change than developed countries.' Evaluate this statement.	[13]
	Candidates could argue in support of the statement by emphasizing that less developed countries (LDCs) are generally more vulnerable to the impacts of climate change due to a convergence of structural and socioeconomic factors. LDCs' economies often depend heavily on climate-sensitive sectors like agriculture and tourism, making them especially susceptible to droughts, floods, and temperature anomalies that disrupt livelihoods and food security. Candidates could note that these countries usually have weaker institutions, limited infrastructure, and far fewer resources to finance adaptation or recovery, compounding their vulnerability when disasters strike. Furthermore, candidates could highlight that, while developed nations historically contribute far more to carbon emissions, LDCs disproportionately suffer severe consequences such as intensified poverty, internal displacement, and resource insecurity. This cycle of vulnerability is difficult to break without significant outside assistance or investment. By contrast, developed nations typically possess more robust social safety nets, diversified economies, and greater adaptive capacity, allowing them to manage and recover from climate shocks more effectively and equitably. *Levels marked using Generic Level Descriptors for H1 essays*	
6	Evaluate the view that the state is the principal actor in addressing climate change.	[13]
	Candidates could adopt the approach that while the state has long been regarded as the principal actor in addressing climate change—playing a key role by setting national policy frameworks, implementing emissions targets, and negotiating international agreements—this perspective is increasingly seen as incomplete. Candidates could argue that	

	governments are limited by political cycles, vested interests, and competing national priorities, which can constrain ambitious action. Therefore, effective climate action now requires a multi-stakeholder strategy: non-state actors such as transnational corporations can drive technological innovation and decarbonize supply chains, civil society groups can hold policymakers accountable and mobilize support, and sub-national governments (like states and cities) often lead on local adaptation and mitigation. Candidates should consider framing the state as a necessary coordinator and enabler, but not as the sole principal actor in climate change mitigation and adaptation. This broader perspective better reflects the complexity, scale, and collaborative nature of successful contemporary climate action. *Levels marked using Generic Level Descriptors for H1 essays*	